

Honors in Physics Research Report

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Reaserch Project Title: Universality class of control transition in
Heizenberg Spin Chain

I have emailed the final version of this report to my mentor: We are studying a phase transition between overall chaotic and controlled dynamics in a controlled Heizenberg spin chain. We have coompleted the process of finding the critical exponents of the phase transition. Though we expected the transition to be of the Directed Percolation (DP) universality class, the critical exponents we found differ from those of (DP). This was suprising as we had good reason to expect our transition to be DP. We recall now with seriousness our hamiltonian

$$H = \sum_j (-\vec{J}(t) \circ \vec{S}_{j+1}) \cdot \vec{S}_j, \quad (0.1)$$

in particular emphasizing that we made the J_x, J_y coupling constants random in time (this was done to eliminate the presense of solitons that appeared in the spin chain). This time randomness is a none trivial pertubation in the sense of the Renormalization Group. It follows that it may be the reason our phase transition does not excibit the know critical values of DP. We are now seeking to dertermine if our phase transition belongs to a so called “Time Random Directed Percolation” universality class. To do this we are preturbing (randomly in time) the control parameter of Stavskaya’s probabilistic cellular automaton (STPCA). STPCA is known to be of the DP universality class. If the critical exponents of our random in time version of STPCA agree with the ones of our controlled spin chain, it will give us good reason to expect that the phase transition we are studing belongs to a “Time Random DP” universality class. Finding a way to preturb STPCA to create time randomness similar to the kind in our spin chain (and such that the phase transition is preserved) has proven to be a challange, and one that is the subject of present reaserch. In the end we plan to submit a manuscript of this work before the summer and we expect a publication.